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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Traiblazers
Selected Domain	Domain 1: The Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4: Quality Education
Project / Solution Name	SkillTide
Name (Leader)	HUANG BAONING
Email & Phone (Leader)	SWE2609018@xmu.edu.my · +60 1123419021
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing the tasks inside entry-level technical jobs faster than degree programmes and career advice can follow. In Malaysia, 56% of working professionals already use AI tools at work, but only 26% have been trained to use them [1]. TalentCorp expects about 697,000 workers in 22 sectors to be significantly affected by AI, digitalisation and the green economy within three to five years [2]; its first study counted 37,000 such jobs in electrical and electronics (E&E) and 30,000 in ICT [3]. Meanwhile, online job offerings fell 11.4% in one quarter, to 555,509 in Q2 2026 [4].

Who is affected. Final-year students and fresh graduates in computer science, E&E and mechatronics, who are about to enter exactly these sectors. Before the job hunt they must choose a first role and what to learn, and they face three gaps:

- **No live view of change:** skill and job lists are static and out of date by the time they are published.
- **No evidence behind advice:** a general chatbot cites nothing and can invent jobs, courses or numbers.
- **No plan that fits final year:** students have a few hours a week, little money and a fixed deadline.

Why it matters. Graduates who train for tasks AI is taking over start their careers in a tighter market, E&E and ICT employers keep their skill gaps, and students without good career advice fall further behind. This works against SDG target 4.4: more young people with relevant skills for decent jobs.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillTide is a web app that shows a final-year student which technical roles are rising or under pressure as AI spreads, where they fit, and which free or low-cost courses close their gaps. The prototype is live at skilltide.replit.app.

How it applies AI. SkillTide forecasts AI's effect task by task, not from job titles. For 657 of the 658 O*NET tasks behind 30 roles, it combines measured AI use (Anthropic Economic Index) with a published ceiling on how far a language model can speed the task up (Eloundou et al.), then splits exposure into work AI is likely to assist and work it is likely to automate. Hiring demand (Indeed

postings in the US, UK and Australia) and Malaysia's Critical Occupations List adjust each role's outlook. Two AI-adoption scenarios run side by side, and six dated data snapshots can be replayed.

What the student does. Profile (a CV read in the browser, a skills form or quick questions), Future Skills Radar, Matches (a visible score with weights the student sets) and Pathway (a week-by-week plan ending in a portfolio project). A Guide answers “why?” and “what if?” with sources. Marking a step done raises skill levels, earns badges and re-plans; study circles connect students who share gaps; the interface is in English, Bahasa Melayu and Simplified Chinese.

Workshop tools we build on:

- **GenAI API:** an optional OpenAI model (gpt-6-luna) rewords the Guide's answers; a grounding check rejects any answer with numbers not in our data and falls back to the rule-based answer.
- **Database architecture:** a versioned dataset built from 13 cited sources by a tested pipeline; in Phase 2, Replit PostgreSQL for anonymous pilot metrics.
- **Cloud hosting:** the site and API run on Replit Autoscale; if the API is unreachable, the browser runs the same engine.

3. Technical Architecture & Implementation Plan

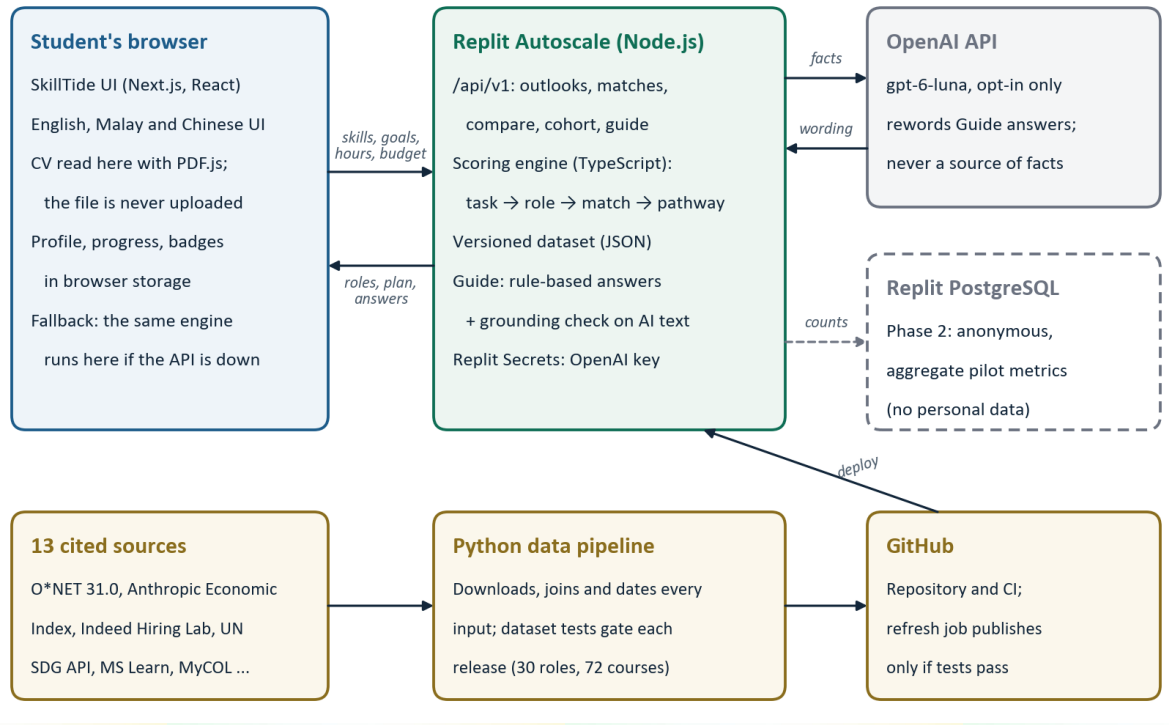
3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

One TypeScript code base: the same scoring engine runs on the server and, as a fallback, in the browser. Code: github.com/ArcueidMP/skilltide.

System block diagram



- **Front end:** Next.js 16 (static export), React 19, TypeScript 5, Tailwind CSS 4 and Recharts. Mozilla PDF.js reads a CV inside the browser. The interface is translated into English, Bahasa Melayu and Simplified Chinese.
- **Back end / API:** a versioned JSON API (/api/v1: outlooks, matches, scenario comparison, cohort, guide) in one request handler, served by a Node.js server on Replit Autoscale; the same handler also builds as a Cloudflare Worker. Requests carry only skill levels, goals, study hours and budget, never the CV, and the server stores no personal data.
- **Scoring engine:** pure TypeScript functions (task exposure → role outlook → match score → pathway), checked by 106 Vitest tests and a 12-case synthetic evaluation.

- **GenAI:** the OpenAI API with gpt-6-luna, behind an opt-in "AI wording" switch for the Guide only. The model receives the facts SkillTide computed; if its answer contains a number that is not in those facts, the Guide shows the rule-based answer instead. Answers are cached for a week and rate-limited.
- **Data layer:** a Python (standard library) pipeline downloads public data and builds one versioned dataset (30 roles, 57 skills, 72 courses, 6 dated snapshots). Dataset tests gate every release, and a GitHub Actions refresh job publishes only when they pass. Profiles and progress stay in the student's browser storage. Phase 2: Replit PostgreSQL for anonymous, aggregate pilot metrics.
- **Data sources:** O*NET 31.0, Anthropic Economic Index, GPTs-are-GPTs exposure labels, Indeed Hiring Lab, Hacker News hiring threads, UN SDG API, Microsoft Learn Catalog API and MyCOL (TalentCorp).
- **Tooling:** GitHub (version control and CI), Replit (build and deployment), Vitest and ESLint.

Sponsor platform: yes, Replit. The GitHub repository is imported into a Repl, which builds the site and runs the API server. It is published as a Replit Autoscale deployment (region Asia) at skilltide.replit.app, and the OpenAI key is kept in Replit Secrets. Repl: @ArcueidMP/skilltide. We did not use a sponsor-provided GenAI API: the optional AI wording uses our own OpenAI API key.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise. Our brainstorm started from final-year students in CS, E&E and mechatronics: they must pick a first role with a few hours a week and little money, and generic advice gives them nothing they can check.

Define. How might a final-year student see which roles AI is lifting or pressuring, where they fit and what to learn next, with evidence they can check?

Ideate. We rejected "upload a CV and ask a chatbot" because it cites nothing. We chose task-level formulas with visible weights, labelled assumptions and two scenarios, and allowed a language model only to reword.

Prototype and test. We shipped a working app early and iterated: teammates asked for badges, study groups and Malay and Chinese, and all three are built. 106 automated tests run in CI, and a 12-case synthetic evaluation checks rankings, budgets and fairness.

Deployment. The app already runs on Replit Autoscale. A failed data refresh never publishes, the browser runs the engine if the API fails, and without an AI key the Guide answers with rules. Phase 2 adds verified courses for the open gaps and a test with final-year volunteers.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 4.4:** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." [5] SkillTide also links each role to 11 more targets in SDGs 7, 8, 9 and 12, and each link names the O*NET task that triggered it.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	• Only **26%** of Malaysian professionals have been trained to use AI tools, although **56%** already use them at work [1]. • **37,000** E&E jobs and **30,000** ICT jobs are significantly affected by AI and digitalisation within three to five years [3], part of about **697,000** workers in 22 sectors [2]. • **Measured in SkillTide:** across our 180-student synthetic cohort (60 each in CS, E&E and mechatronics; data of 16 Oct 2026, Moderate AI scenario), verified free or low-cost courses close **78%** of the priority skill gaps for each student's top role (696 of 889), and only **28%** of students (51 of 180) can close all of them [6].
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• **Gap coverage:** % of priority skill gaps on students' pathways that a verified free or low-cost course closes, measured on the synthetic cohort with npm run impact. • **Fully covered students:** % of cohort students who can close every priority gap. • **Pilot usability:** % of final-year volunteers who reach a first pathway in under 5 minutes and can explain why their top role ranks first.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• Gap coverage from **78% to at least 90%**, and fully covered students from **28% to at least 50%**. Most open gaps sit in three skills with no verified free course yet (equipment maintenance, lean and quality, MATLAB), so we will verify and add free courses for them. • **At least 80%** of 10 or more final-year volunteers reach a first pathway in under 5 minutes and can explain their top role's score (first measured in Phase 2).

Sources for sections 2 and 4

[1] Hays Malaysia, press release on the Hays Asia Salary Guide 2025, 7 Mar 2025. hays.com.my/press-release/content/future-of-work-malaysia-technology-ai-impact

[2] Bernama, TalentCorp Group CEO on the updated AI and digital impact study, 1 May 2026. bernama.com/en/general/news.php?id=2551724

[3] The Star, Human Resources Minister on the TalentCorp Phase 1 impact study, 19 Aug 2025. thestar.com.my/news/nation/2025/08/19/ai-digitalisation-and-green-tech-to-affect-over-620000-jobs-in-five-years-says-sim

[4] Department of Statistics Malaysia, Big Data Analytics: Job Market Insights Q2 2026, 18 Aug 2026. dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-q22026

[5] United Nations, Goal 4 targets (sdgs.un.org/goals/goal4); wording read through the UN SDG API on 21 Oct 2026.

[6] SkillTide engine on its 16 Oct 2026 dataset; reproduce with npm run impact in github.com/ArcueidMP/skilltide.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js, React, React DOM	Code (web framework)	MIT
Tailwind CSS, tailwind-merge, clsx	Code (styling)	MIT
Recharts, Motion	Code (charts, animation)	MIT
Lucide icons (lucide-react)	Code (icons)	ISC
TypeScript	Code (language, compiler)	Apache-2.0
Mozilla PDF.js (pdfjs-dist)	Code (reads CV text in the browser)	Apache-2.0
Vitest, tsx, ESLint, eslint-config-next	Code (development tools)	MIT
Wrangler (Cloudflare)	Code (deployment tool)	MIT or Apache-2.0
IBM Plex Sans, IBM Plex Mono, Source Serif 4 (Fontsource)	Code (web fonts)	SIL Open Font License 1.1

Name	Type (code/dataset/model)	License
O*NET 31.0 Database (USDOL/ETA)	Dataset (roles, tasks, skills)	CC BY 4.0
Anthropic Economic Index; Anthropic robot exposure index	Dataset (observed AI use per task)	CC BY (data)
GPTs are GPTs exposure labels (Eloundou et al.)	Dataset (task exposure ceiling)	MIT
Indeed Hiring Lab Job Postings Index and AI Tracker	Dataset (hiring demand)	CC BY 4.0
Hacker News “Who is hiring?” threads (Algolia HN API)	Dataset (skill mention counts only)	Public API; post text not stored
UN SDG Global Database API	Dataset (goal and target wording)	UN terms of use; cited
Microsoft Learn Catalog API	Dataset (course links and durations)	Microsoft APIs Terms of Use
MyCOL 2024/2025 (TalentCorp); new-occupation lists (TalentCorp, China MOHRSS, WEF)	Dataset (cited facts)	Public reports; cited with links
Malaysian context figures (Hays, Bernama, The Star, DOSM, PwC, HRD Corp)	Dataset (cited figures shown as context)	Public sources; cited with links
Course provider pages for 55 courses (Cisco, Kaggle, MathWorks, HRD Corp e-LATiH and others)	Dataset (hours and prices quoted with links)	Facts cited; no course content copied
OpenAI gpt-6-luna (OpenAI API)	Model (optional Guide wording)	OpenAI API terms (proprietary)
Claude Code (Anthropic), Codex (OpenAI)	Model (AI coding assistants used during development)	Each provider's commercial terms (proprietary)

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



HUANG BAONING, Team Leader

Date: 24 October 2026